

The Financial Economics and Spectral Geometry of Coinage

A Dynamic Theory of Constant-Width Monetary Instruments

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Abstract

This manuscript develops a general theory of coinage in which the geometry of money is treated as an endogenous economic design variable. The theory includes the circle as a special case but permits arbitrary convex curves of constant width, truncated spectrally into odd Fourier harmonics. A coin is characterized by face value, outer width, thickness, material density, spectral geometry, topology, production technology, holding cost, monetary opportunity cost, and its position within an installed stock of circulating currency.

The analysis proceeds through five stages. First, a single denomination is optimized over a finite-dimensional space of constant-width curves and optional central perforation. Second, spectral and topological phase diagrams are constructed as functions of material prices, interest rates, monetary demand, and manufacturing technology. Third, a complete denomination system is treated as a portfolio whose components must be jointly optimized because geometric differentiation reduces recognition and confusion costs. Fourth, the installed stock of money introduces dynamic adjustment costs, hysteresis, and optimal control. Fifth, an empirical state-space and structural estimation framework is specified.

The framework connects geometry, mechanics, metallurgy, production economics, finance, monetary economics, psychology, human recognition, accessibility, industrial engineering, statistical inference, algorithms, artificial intelligence, political economy, international monetary relations, welfare economics, strategic resilience, and institutional design. A central conclusion is that the circle is not theoretically privileged: it is the zero-spectrum, zero-hole phase of a much larger space of feasible monetary geometries.

The paper ends with “The End”

Contents

1	Introduction	3
2	Scope and Interdisciplinary Structure	4
3	Geometry of Constant-Width Coinage	4
3.1	Support functions	4
4	Finite Spectral Representation	5
5	Convexity and Curvature	6
6	Area, Density, and Mass	6
7	Mechanics and Physics	7
8	Chemistry and Materials Science	8
9	Production Cost and Industrial Economics	8
10	Spectral Activation	9

11 Topology: Solid and Perforated Money	10
12 Finance and the Economics of Negative Interest Rates	10
13 Welfare and the Static Optimization Problem	11
14 Spectral and Topological Phases	12
15 Phase Interactions	12
16 Psychology, Recognition, and Human Factors	13
17 Multi-Denomination Coinage as Portfolio Theory	13
18 Political Economy and Institutional Design	14
19 International Relations and Monetary Resilience	14
20 Dynamic Installed-Stock Coinage	15
21 Hysteresis	15
22 A Dynamic Welfare Objective	16
23 Statistics and Empirical Calibration	16
24 Identification	17
25 State-Space Representation	17
26 Computer Science and Algorithms	18
26.1 Pseudo-code	18
27 Rotational Symmetry	19
28 Data Science and AI/ML	19
29 Philosophical Interpretation	20
30 Religion, Culture, and Symbolic Constraints	20
31 Strategic and Warfare Economics Interpretation	20
32 Vector Graphics	21
32.1 Constant-width support geometry	21
32.2 Spectral phase space	21
32.3 Dynamic coinage architecture	21
33 Five-Stage Research Program	22
34 Main Predictions	22
35 Limitations	22
36 Conclusion	23

List of Figures

1	Schematic representation of a non-circular constant-width coin.	21
2	Illustrative spectral-topological phase transitions.	21
3	Dynamic architecture of the theory of coinage.	21

List of Tables

1	Disciplines entering the theory of coinage	4
2	Benchmark harmonic manufacturing costs	9
3	Benchmark spectral activation thresholds	9
4	Five stages of the theory	22

1 Introduction

A coin is simultaneously a geometric body, a manufactured object, a unit of account, a medium of exchange, a store of value, a public infrastructure component, and an element of state capacity. Traditional treatments of coinage usually take its geometry as exogenous. A circular coin is assumed, and the economic questions concern denomination, material composition, seigniorage, production cost, circulation, or withdrawal.

This manuscript reverses that assumption.

The fundamental proposition is that the physical geometry of money can itself be treated as an economic control variable.

Let $W > 0$ denote the outer width of a coin. Let the coin belong to the class of convex planar bodies of constant width W . The circle is one member of this class, but not the only member. Non-circular constant-width curves provide a continuous family of alternatives possessing the same width in every direction while differing in:

- area;
- mass;
- curvature;
- boundary length;
- tactile characteristics;
- stacking and handling properties;
- manufacturing complexity;
- recognizability;
- compatibility with machines;
- holding cost;
- counterfeit resistance;
- strategic supply requirements.

Let F denote face value. Let the total cost of producing a coin be

$$c(F, t) = f(F) + v(F, t),$$

where $f(F)$ denotes fixed denomination-specific cost and $v(F, t)$ denotes variable cost.

Let $r(t)$ denote the nominal bank interest rate. Importantly,

$$r(t) < 0$$

is permitted.

Negative interest rates alter the economics of physical currency because cash has a nominal return near zero. The opportunity cost of holding currency may therefore become negative, increasing the attractiveness of physical monetary instruments. Coin geometry can then affect not only manufacturing economics but the transmission of monetary policy.

The resulting research question is:

What geometry, topology, material content, and denomination structure should a monetary authority choose when money itself is treated as a dynamically optimized physical technology?

2 Scope and Interdisciplinary Structure

The theory is deliberately multidisciplinary.

Table 1: Disciplines entering the theory of coinage

Field	Role in the theory
Geometry	Constant width, support functions, curvature, area, topology.
Mechanics	Mass, inertia, handling, wear, stress, impact, rolling.
Physics	Density, friction, acoustic response, electromagnetic properties.
Chemistry	Alloy composition, corrosion, oxidation, plating, material degradation.
Economics	Production, resource allocation, seigniorage, welfare.
Finance	Opportunity cost, discounting, interest-rate sensitivity.
Monetary economics	Currency demand, effective lower bounds, denomination systems.
Psychology	Recognition, tactile discrimination, cognitive confusion.
Psychiatry	Human behavioural heterogeneity and the limits of purely representative-agent assumptions.
Biology	Human sensory and motor constraints; wear and environmental interaction.
Political science	Institutional authority over money and public legitimacy.
International relations	Currency recognition, sanctions resilience, reserve and settlement systems.
Diplomacy	Monetary symbolism and interoperability across jurisdictions.
Welfare economics	Social optimization under production and recognition externalities.
Warfare economics	Strategic material conservation and monetary continuity under disruption.
Statistics	Estimation, inference, causal identification, uncertainty quantification.
Computer science	Optimization, computational geometry, data structures, simulation.
Data science	Historical coin databases, manufacturing records, circulation data.
AI/ML	Surrogate optimization, phase classification, state estimation, anomaly detection.
Philosophy	Ontology of money and the distinction between nominal and physical value.
Religion and culture	Symbolic design constraints, iconography, public acceptability.

The inclusion of these fields does not imply that every coinage problem requires equal use of every discipline. Rather, coinage is an intrinsically multi-constraint design problem, and different jurisdictions may attach different weights to different objectives.

3 Geometry of Constant-Width Coinage

3.1 Support functions

Let $K \subset \mathbb{R}^2$ be a compact convex body. Define its support function by

$$h(\theta) = \max_{x \in K} x \cdot (\cos \theta, \sin \theta).$$

The body has constant width W if

$$h(\theta) + h(\theta + \pi) = W$$

for every θ .

Define

$$R = \frac{W}{2}.$$

The circle of radius R is represented by

$$h(\theta) = R.$$

A general constant-width support function can be written as

$$h(\theta) = R + g(\theta),$$

where

$$g(\theta + \pi) = -g(\theta).$$

Hence only odd Fourier harmonics are permitted.

4 Finite Spectral Representation

Let

$$\mathcal{N} = \{3, 5, 7, 9\}.$$

The first, translational harmonic $n = 1$ is excluded by fixing the origin. Define

$$h(\theta) = R \left[1 + \sum_{n \in \mathcal{N}} \frac{x_n \cos(n\theta) + y_n \sin(n\theta)}{2n} \right].$$

Equivalently,

$$x_n = z_n \cos \phi_n, \quad y_n = z_n \sin \phi_n,$$

so that

$$z_n^2 = x_n^2 + y_n^2.$$

The geometric state vector is

$$\mathbf{q} = (x_3, y_3, x_5, y_5, x_7, y_7, x_9, y_9, \lambda).$$

Here λ controls the central hole.

Theorem 1 (Automatic constant width). *Every support function of the above spectral form satisfies*

$$h(\theta) + h(\theta + \pi) = W.$$

Proof. For odd n ,

$$\cos(n(\theta + \pi)) = -\cos(n\theta),$$

and

$$\sin(n(\theta + \pi)) = -\sin(n\theta).$$

Hence all spectral terms cancel when $h(\theta)$ and $h(\theta + \pi)$ are added:

$$h(\theta) + h(\theta + \pi) = 2R = W.$$

□

Thus constant width is embedded directly into the parameterization rather than being imposed numerically.

5 Convexity and Curvature

For a smooth support function, the radius of curvature is

$$\mathcal{R}(\theta) = h(\theta) + h''(\theta).$$

Therefore

$$\frac{\mathcal{R}(\theta)}{R} = 1 - \sum_{n \in \mathcal{N}} \beta_n [x_n \cos(n\theta) + y_n \sin(n\theta)],$$

where

$$\beta_n = \frac{n^2 - 1}{2n}.$$

The relevant coefficients are

$$\beta_3 = \frac{4}{3}, \quad \beta_5 = \frac{12}{5}, \quad \beta_7 = \frac{24}{7}, \quad \beta_9 = \frac{40}{9}.$$

Convexity requires

$$\mathcal{R}(\theta) \geq 0 \quad \forall \theta.$$

For numerical implementation, define

$$\theta_k = \frac{2\pi k}{K}, \quad k = 0, \dots, K-1.$$

The optimization constraints are

$$1 - \sum_{n \in \mathcal{N}} \beta_n [x_n \cos(n\theta_k) + y_n \sin(n\theta_k)] \geq \varepsilon_{\mathcal{R}}.$$

A dense validation grid must subsequently verify continuous feasibility.

Remark 2. Grid feasibility is exact at the selected grid points but is not logically equivalent to continuous convexity. Therefore a production implementation should use refinement, certified trigonometric positivity methods, or conservative bounds where necessary.

6 Area, Density, and Mass

For a constant-width body represented by the support function above, the outer area is

$$A_{\text{out}} = \frac{1}{2} \int_0^{2\pi} (h(\theta)^2 - h'(\theta)^2) \, d\theta.$$

Orthogonality yields

$$A_{\text{out}} = \pi R^2 \left[1 - \sum_{n \in \mathcal{N}} \alpha_n z_n^2 \right],$$

where

$$\alpha_n = \frac{n^2 - 1}{8n^2}.$$

Hence

$$\alpha_3 = \frac{1}{9}, \quad \alpha_5 = \frac{3}{25}, \quad \alpha_7 = \frac{6}{49}, \quad \alpha_9 = \frac{10}{81}.$$

Suppose the inner hole has radius

$$r_H = \lambda R, \quad 0 \leq \lambda < 1.$$

Its area is

$$A_H = \pi \lambda^2 R^2.$$

Therefore material area is

$$A_{\text{mat}} = \pi R^2 m(\mathbf{z}, \lambda),$$

where

$$m(\mathbf{z}, \lambda) = 1 - \lambda^2 - \frac{1}{9}z_3^2 - \frac{3}{25}z_5^2 - \frac{6}{49}z_7^2 - \frac{10}{81}z_9^2.$$

Let $\rho(\omega)$ denote the material density as a function of radial coordinate ω . In the simplest homogeneous model,

$$\rho(\omega) = \begin{cases} 0, & 0 \leq \omega \leq w, \\ \rho > 0, & w < \omega \leq W. \end{cases}$$

If thickness is τ , homogeneous mass is

$$M = \rho \tau \pi R^2 m.$$

For radial density variation, the mass functional is more generally

$$M = \tau \int_K \rho(\|\mathbf{x}\|) dA.$$

Thus density heterogeneity, plating, alloy gradients, and composite construction can be incorporated without changing the geometric framework.

7 Mechanics and Physics

A coin is a physical body subject to impact, friction, wear, vibration, and machine interaction.

For a homogeneous planar lamina of thickness τ , the moment of inertia about the central axis is

$$I_z = \rho \tau \int_{K \setminus H} (x^2 + y^2) dA.$$

The boundary parameterization under normal angle θ is

$$\mathbf{r}(\theta) = h(\theta)\mathbf{n}(\theta) + h'(\theta)\mathbf{t}(\theta),$$

where

$$\mathbf{n}(\theta) = (\cos \theta, \sin \theta),$$

and

$$\mathbf{t}(\theta) = (-\sin \theta, \cos \theta).$$

Consequently,

$$\|\mathbf{r}(\theta)\|^2 = h(\theta)^2 + h'(\theta)^2.$$

The minimum radial distance is

$$r_{\min} = \min_{\theta} \sqrt{h(\theta)^2 + h'(\theta)^2}.$$

A centered circular hole is feasible when

$$\boxed{\lambda R \leq r_{\min} - \varepsilon_H.}$$

The hole is therefore not a purely topological variable; it interacts directly with the spectral geometry. Mechanical engineering introduces additional constraints, including:

- minimum ligament thickness;
- resistance to deformation;
- fatigue under repeated machine handling;
- edge damage;
- friction coefficients;
- acoustic and electromagnetic signatures.

These can be represented by constraint functions

$$G_j(\mathbf{q}) \geq 0, \quad j = 1, \dots, J.$$

8 Chemistry and Materials Science

Let the coin be composed of L materials with mass fractions

$$\omega_\ell, \quad \sum_{\ell=1}^L \omega_\ell = 1.$$

A first-order effective density is

$$\rho_{\text{eff}} = \left(\sum_{\ell=1}^L \frac{\omega_\ell}{\rho_\ell} \right)^{-1}.$$

Material cost per unit mass is

$$p_t = \sum_{\ell=1}^L \omega_\ell p_{\ell,t}.$$

The material component of production cost is

$$p_t M.$$

Corrosion and chemical degradation can be represented by a state variable χ_t :

$$\chi_{t+1} = \chi_t + g(\chi_t, T_t, H_t^{\text{env}}, S_t^{\text{chem}}) + u_t^\chi.$$

Here T_t denotes temperature and H_t^{env} denotes environmental humidity.

Geometry can influence chemical degradation through exposed surface area and edge complexity.

9 Production Cost and Industrial Economics

Let F denote face value. The general production cost is

$$c(F, t) = f(F) + v(F, t).$$

For a spectral coin, normalize cost by an appropriate monetary scale and define

$$\begin{aligned} \hat{c}(\mathbf{q}) = & \hat{c}_0 + \hat{p} m(\mathbf{z}, \lambda) + \mathbf{1}_{\{\lambda > 0\}} \hat{K}_H + \hat{\eta} \lambda + \frac{\hat{\chi}}{2} \lambda^2 + \frac{\hat{\psi}}{4} \lambda^4 \\ & + \sum_{n \in \mathcal{N}} \left[\frac{\kappa_n}{2} z_n^2 + \frac{\gamma_n}{4} z_n^4 \right] + \frac{1}{2} \sum_{m < n} \xi_{mn} z_m^2 z_n^2. \end{aligned} \quad (1)$$

The interpretation is:

- $\hat{p}$: normalized material price;
- $\hat{K}_H$: fixed cost of introducing a hole;
- $\hat{\eta}, \hat{\chi}, \hat{\psi}$: hole-processing technology;
- κ_n : quadratic complexity cost;
- γ_n : nonlinear stabilization;
- ξ_{mn} : cross-mode complementarity or competition.

A benchmark assumption is

$$\kappa_n = \kappa_0 + \kappa_1 n^2,$$

with

$$\kappa_0 = 0.10, \quad \kappa_1 = 0.015.$$

Then

Table 2: Benchmark harmonic manufacturing costs

n	3	5	7	9
κ_n	0.235	0.475	0.835	1.315

10 Spectral Activation

Ignoring higher-order interactions, the contribution of mode n to the objective contains the material-saving term

$$-\hat{p}\alpha_n z_n^2$$

inside cost and the manufacturing penalty

$$\frac{\kappa_n}{2} z_n^2.$$

The local activation threshold is therefore

$$\hat{p}_n^{\text{crit}} = \frac{4n^2}{n^2 - 1} \kappa_n.$$

Under the benchmark:

Table 3: Benchmark spectral activation thresholds

Mode	3	5	7	9
$\hat{p}_n^{\text{crit}}$	1.0575	1.9792	3.4096	5.3258

Proposition 3 (Low-frequency priority). *Suppose κ_n grows sufficiently rapidly with n while material savings are governed by α_n . Then the first spectral deformation away from the circle is generically associated with the lowest economically feasible odd harmonic.*

Proof. The circle corresponds to $z_n = 0$ for all n . The quadratic coefficient governing local activation is proportional to

$$\frac{\kappa_n}{2} - \hat{p}\alpha_n.$$

A mode becomes locally attractive when this coefficient changes sign. The first mode to activate minimizes

$$\frac{\kappa_n}{2\alpha_n} = \frac{4n^2}{n^2 - 1} \kappa_n.$$

If this expression is increasing in n , the lowest admissible harmonic is the first to activate. In the present representation, that harmonic is $n = 3$. $\square$

11 Topology: Solid and Perforated Money

Define topology

$$\mathcal{T} \in \{0, 1\}.$$

If

$$\mathcal{T} = 0,$$

then

$$\lambda = 0.$$

If

$$\mathcal{T} = 1,$$

then

$$\lambda > 0.$$

A hole has several simultaneous effects:

- reduces material use;
- may alter tactile recognition;
- creates additional tooling requirements;
- changes stiffness and mechanical durability;
- changes machine sensing;
- may affect cash holding and portability.

Thus the hole is a discrete technological regime rather than merely a continuous change in area.

12 Finance and the Economics of Negative Interest Rates

Let $r(t)$ denote the bank interest rate.

The theory explicitly allows

$$\boxed{r(t) < 0.}$$

Let $\hat{H}(\mathbf{q})$ denote geometry-dependent physical holding cost:

$$\begin{aligned} \hat{H}(\mathbf{q}) &= \hat{h}_0 + \hat{h}_M m(\mathbf{z}, \lambda) + \hat{h}_H(\lambda) \\ &\quad + \sum_{n \in \mathcal{N}} q_n z_n^2 + \frac{1}{2} \sum_{m < n} v_{mn} z_m^2 z_n^2. \end{aligned} \tag{2}$$

A reduced-form cash-demand equation is

$$\boxed{n_c(\mathbf{q}; r) = \frac{1}{1 + \exp[-\alpha - \beta(-\hat{H}(\mathbf{q}) - r)]}.$$

Differentiation gives

$$\frac{\partial n_c}{\partial r} = -\beta n_c(1 - n_c),$$

while

$$\frac{\partial n_c}{\partial(-\hat{H})} = \beta n_c(1 - n_c).$$

Therefore the geometry-specific monetary substitution threshold is

$$r^\dagger(\mathbf{q}) = -\hat{H}(\mathbf{q}) + \frac{\alpha}{\beta}.$$

Proposition 4 (Monetary geometry window). *The marginal effect of a reduction in physical holding cost on cash demand is maximized when $n_c = 1/2$.*

Proof. The marginal effect is

$$\beta n_c(1 - n_c).$$

Differentiating with respect to n_c gives

$$\beta(1 - 2n_c) = 0,$$

which implies

$$n_c = \frac{1}{2}.$$

The second derivative is $-2\beta < 0$ when $\beta > 0$. □

Hence the largest marginal effect of coin geometry need not occur at the most negative interest rate. At sufficiently negative rates,

$$n_c \rightarrow 1,$$

and

$$n_c(1 - n_c) \rightarrow 0.$$

Cash demand saturates.

13 Welfare and the Static Optimization Problem

Let the normalized authority objective be

$$\hat{J}(\mathbf{q}) = [1 - \hat{c}(\mathbf{q})]n_c(\mathbf{q}; r) - \frac{\hat{\omega}}{2}n_c(\mathbf{q}; r)^2.$$

The authority solves

$$\begin{aligned} & \max_{\mathbf{q}} \quad \hat{J}(\mathbf{q}) \\ & \text{subject to} \quad \mathcal{R}(\theta) \geq 0, \quad \forall \theta, \\ & \quad \quad \quad 0 \leq \lambda \leq \frac{r_{\min}}{R} - \varepsilon_H, \\ & \quad \quad \quad G_j(\mathbf{q}) \geq 0, \quad j = 1, \dots, J. \end{aligned} \tag{3}$$

The objective can be extended to include:

- environmental externalities;

- accessibility;
- counterfeit losses;
- machine compatibility;
- strategic material risk;
- distributional welfare.

14 Spectral and Topological Phases

Define the active spectral set

$$\mathcal{A} = \{n : z_n > z_{\text{tol}}\}.$$

Define the phase

$$\Pi = (\mathcal{A}, \mathcal{T}).$$

Because

$$\mathcal{A} \subseteq \{3, 5, 7, 9\},$$

there are

$$2^4 = 16$$

spectral families.

Each family has two topological states, so the first model contains

$$32$$

basic spectral-topological families.

For each phase,

$$V_{\Pi}(\hat{p}, r) = \max_{\mathbf{q} \in \mathfrak{G}_{\Pi}} \hat{J}(\mathbf{q}; \hat{p}, r).$$

The optimal phase is

$$\Pi^*(\hat{p}, r) = \arg \max_{\Pi} V_{\Pi}(\hat{p}, r).$$

The boundary between two phases satisfies

$$V_{\Pi_1}(\hat{p}, r) = V_{\Pi_2}(\hat{p}, r).$$

15 Phase Interactions

Cross-mode interaction determines whether optimal geometry is sparse or hybrid.

If

$$\xi_{mn} > 0,$$

simultaneous activation is penalized.

If

$$\xi_{mn} < 0,$$

simultaneous activation is rewarded.

Thus positive interactions may produce

$$\emptyset \rightarrow \{3\} \rightarrow \{5\} \rightarrow \{7\},$$

while negative interactions may produce

$$\emptyset \rightarrow \{3\} \rightarrow \{3, 5\} \rightarrow \{3, 5, 7\}.$$

This is analogous to substitution and complementarity in economics and to interaction effects in statistical physics.

16 Psychology, Recognition, and Human Factors

Coins are identified through visual, tactile, auditory, and contextual signals.

Let D_{ij}^{spec} denote a spectral distance between denominations:

$$D_{ij}^{\text{spec}} = \sum_n w_n [(x_{n,i} - x_{n,j})^2 + (y_{n,i} - y_{n,j})^2] + w_\lambda (\lambda_i - \lambda_j)^2.$$

A reduced-form confusion probability is

$$P_{ij}^{\text{conf}} = \exp[-\zeta D_{ij}^{\text{spec}}].$$

Expected confusion loss can then be written

$$C_{ij}^{\text{conf}} = L_{ij} P_{ij}^{\text{conf}}.$$

The theory can incorporate heterogeneous users.

Let a index population groups. Then

$$P_{ij,a}^{\text{conf}} = \exp[-\zeta_a D_{ij}^{\text{spec}}].$$

This permits accessibility constraints for users with different visual, tactile, motor, or cognitive characteristics.

The relevant principle is not that any particular medical or psychological condition determines monetary design. Rather, public infrastructure should be robust to human heterogeneity.

17 Multi-Denomination Coinage as Portfolio Theory

Let

$$F_1 < F_2 < \dots < F_D$$

denote denominations.

Each denomination has geometry

$$\mathbf{q}_d.$$

Define

$$\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_D).$$

The system objective is

$$\begin{aligned} \mathcal{J}(\mathbf{Q}) = & \sum_{d=1}^D \omega_d \hat{J}_d(\mathbf{q}_d) \\ & - \sum_{i < j} C_{ij}^{\text{conf}} - C_{\text{system}}(\mathbf{Q}). \end{aligned} \tag{4}$$

The optimization problem is

$$\begin{aligned} & \max_{\mathbf{Q}} \quad \mathcal{J}(\mathbf{Q}) \\ & \text{subject to} \quad \mathbf{q}_d \in \mathfrak{G}_d, \\ & \quad \quad \quad D_{ij}^{\text{spec}} \geq \underline{D}_{ij}. \end{aligned} \tag{5}$$

Proposition 5 (Geometric diversification). *Suppose independently optimal denomination geometries are insufficiently separated and confusion costs decrease strictly with spectral distance. Then the systemically optimal geometry vector generally differs from the collection of independently optimal geometries.*

Proof. Suppose the independently optimal solution contains a pair (i, j) with

$$D_{ij}^{\text{spec}} < \underline{D}_{ij}.$$

Then the independently optimal vector is infeasible under the systemic recognition constraint.

Even if the constraint is relaxed and confusion is represented as a strictly decreasing penalty, a perturbation increasing separation reduces confusion cost. The systemic optimum therefore balances private denomination-level objectives against the externality generated by similarity. Unless the independent optimum already satisfies the first-order conditions of the augmented objective, the solutions differ. $\square$

Thus a currency system is a geometric portfolio.

18 Political Economy and Institutional Design

Coinage is an exercise of institutional authority.

Let Θ denote the vector of political and institutional constraints:

$$\Theta = (\text{legitimacy, tender, rules, legal public acceptability, industrial capacity, symbolic constraints}).$$

The feasible design set becomes

$$\mathfrak{G}^\Theta = \mathfrak{G} \cap \mathfrak{I}(\Theta).$$

Political economy therefore affects the feasible set rather than merely the objective function.

A theoretically efficient geometry may be rejected if it:

- lacks public legitimacy;
- cannot be manufactured domestically;
- conflicts with statutory requirements;
- produces unacceptable transition costs;
- creates undesirable distributional consequences.

19 International Relations and Monetary Resilience

A currency system operates within an international environment.

Let $s_{\ell,t}$ denote strategic supply risk for material ℓ . A simple risk-adjusted material cost is

$$\tilde{p}_{\ell,t} = p_{\ell,t} + \varphi_{\ell} s_{\ell,t}.$$

Then

$$\tilde{p}_t = \sum_{\ell} \omega_{\ell} \tilde{p}_{\ell,t}.$$

Geopolitical disruption, sanctions, transport interruptions, and supply concentration can therefore alter optimal geometry indirectly through material economics.

The model should be interpreted defensively: strategic resilience means maintaining continuity of monetary infrastructure under disruption, not optimizing for destructive conflict.

20 Dynamic Installed-Stock Coinage

Static optimization assumes instantaneous replacement.

Let

$$s_{d,\pi,t}$$

denote the circulating stock of denomination d in phase π .

Let

$$u_{d,\pi,t}$$

denote new minting and

$$w_{d,\pi,t}$$

denote withdrawal.

Then

$$s_{d,\pi,t+1} = (1 - \delta_{d,\pi,t})s_{d,\pi,t} + u_{d,\pi,t} - w_{d,\pi,t}.$$

Let

$$\mathbf{S}_t$$

denote the installed stock.

The economic state is

$$\mathbf{X}_t = (\mathbf{p}_t, r_t, \boldsymbol{\rho}_t, \boldsymbol{\kappa}_t, \mathbf{D}_t).$$

The authority solves

$$V(\mathbf{S}_t, \mathbf{X}_t) = \max_{\mathbf{U}_t, \mathbf{W}_t} \left\{ \Pi(\mathbf{S}_t, \mathbf{U}_t, \mathbf{W}_t, \mathbf{X}_t) - C_{\text{transition}} + \delta \mathbb{E}_t[V(\mathbf{S}_{t+1}, \mathbf{X}_{t+1})] \right\}. \quad (6)$$

The optimal policy is

$$\mu^* : (\mathbf{S}_t, \mathbf{X}_t) \mapsto (\mathbf{U}_t^*, \mathbf{W}_t^*).$$

21 Hysteresis

Let π denote the incumbent phase and π' an alternative.

A transition occurs when

$$V_{\pi'} - V_{\pi} > K_{\pi \rightarrow \pi'}.$$

The reverse transition requires

$$V_{\pi} - V_{\pi'} > K_{\pi' \rightarrow \pi}.$$

Therefore, generally,

$$p_{\pi \rightarrow \pi'} \neq p_{\pi' \rightarrow \pi}.$$

The same applies to interest-rate thresholds.

Theorem 6 (Coinage hysteresis). *If transition costs are strictly positive, the dynamic switching boundary between two coinage phases generally differs from the static indifference boundary.*

Proof. The static boundary is defined by

$$V_{\pi'} = V_{\pi}.$$

With positive transition cost, switching requires

$$V_{\pi'} - V_{\pi} = K_{\pi \rightarrow \pi'} > 0.$$

Hence the dynamic switching boundary is displaced relative to the static boundary. If reverse costs differ or if continuation values depend on the installed stock, the forward and reverse boundaries differ, generating hysteresis. $\square$

22 A Dynamic Welfare Objective

The complete authority problem can be represented as

$$\max_{\{Q_t, U_t, W_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \delta^t \left[S_t - C_t^{\text{prod}} - C_t^{\text{hold}} - C_t^{\text{conf}} - C_t^{\text{transition}} - \mathcal{E}_t \right]. \quad (7)$$

The components may represent:

- S_t : monetary and transaction services;
- C_t^{prod} : manufacturing cost;
- C_t^{hold} : physical-currency holding cost;
- C_t^{conf} : recognition and confusion losses;
- $C_t^{\text{transition}}$: redesign costs;
- $\mathcal{E}_t$: environmental and strategic externalities.

23 Statistics and Empirical Calibration

Empirical calibration requires observations on:

- physical dimensions;
- mass;
- material composition;
- production cost;
- minting volume;
- withdrawal rates;
- metal prices;
- interest rates;
- currency circulation;
- design transitions.

A spectral estimate of an observed boundary is

$$\hat{h}(\theta) - \frac{W}{2} = \sum_{n \in \mathcal{N}} [\hat{a}_n \cos(n\theta) + \hat{b}_n \sin(n\theta)] + \varepsilon(\theta).$$

The amplitude estimate is

$$\hat{z}_n = \sqrt{\hat{x}_n^2 + \hat{y}_n^2}.$$

A production-cost equation is

$$\begin{aligned} c_{d,t} = & f_d + p_t M_{d,t} + \sum_n \frac{\kappa_n}{2} z_{d,n,t}^2 \\ & + \sum_n \frac{\gamma_n}{4} z_{d,n,t}^4 + K_H \mathbf{1}_{\{\lambda_d > 0\}} + \varepsilon_{d,t}. \end{aligned} \tag{8}$$

A cash-demand equation is

$$\frac{C_t}{Y_t} = \Lambda[\alpha + \beta(-H_t - r_t) + \mathbf{Z}_t' \boldsymbol{\delta}] + u_t.$$

24 Identification

Geometry is endogenous because monetary authorities choose it.

Therefore naive regressions such as

$$C_t = \alpha + \beta z_{3,t} + u_t$$

cannot generally be interpreted causally.

Useful sources of identification include:

1. unexpected material-price movements;
2. technological changes in minting;
3. institutional redesign mandates;
4. interest-rate policy shocks;
5. exogenous supply disruptions.

A structural model is preferable where sufficient data exist.

25 State-Space Representation

Define the latent state

$$\mathbf{X}_t = (p_t, r_t, \boldsymbol{\kappa}_t, \mathbf{S}_t, \mathbf{D}_t).$$

The transition equation is

$$\mathbf{X}_{t+1} = F(\mathbf{X}_t, \Pi_t) + \mathbf{u}_{t+1}.$$

The observation equation is

$$\mathbf{Y}_t = G(\mathbf{X}_t, \Pi_t) + \boldsymbol{\varepsilon}_t.$$

Observed data may include

$$\mathbf{Y}_t = (M_t, c_t, C_t, \text{mint volume, withdrawal, confusion events}).$$

Estimation methods include:

- maximum likelihood;
- Bayesian inference;
- particle filtering;
- sequential Monte Carlo;
- simulation-based estimation.

26 Computer Science and Algorithms

The finite-dimensional optimization problem is nonlinear and nonconvex.

A robust algorithm should enumerate spectral-topological families and perform multi-start constrained optimization.

26.1 Pseudo-code

INPUT:

```
material-price grid P
interest-rate grid R
spectral families A subset of {3,5,7,9}
topology states T in {solid, perforated}
```

FOR each p in P:

FOR each r in R:

candidate_set = empty

FOR each spectral family A:

FOR each topology T:

construct reduced optimization problem

generate initial points:

```
circle
pure-mode states
hybrid states
random low-amplitude states
boundary-adjacent feasible states
```

FOR each initial point:

solve nonlinear constrained problem

validate:

```
convexity on dense grid
hole containment
mass and area consistency
engineering constraints
```

IF valid:

```
remove rotational duplicates
add solution to candidate_set
```

choose feasible candidate with largest objective

record phase, amplitudes, hole, mass,
holding cost, cash demand, and value

OUTPUT:

spectral phase diagram
topology map
optimal mass surface
optimal hole surface
welfare surface

27 Rotational Symmetry

The representation has rotational redundancy.

A rotation by δ changes phases according to

$$\phi_n \mapsto \phi_n + n\delta.$$

If $z_3 > 0$, impose

$$y_3 = 0, \quad x_3 \geq 0.$$

If $z_3 = 0$ but $z_5 > 0$, use

$$y_5 = 0, \quad x_5 \geq 0.$$

This gauge fixing reduces redundant numerical search.

28 Data Science and AI/ML

Artificial intelligence and machine learning are useful as computational tools, not substitutes for the underlying constraints.

Possible applications include:

- surrogate models for expensive optimization;
- Gaussian-process or neural approximations of $V_{\Pi}(\hat{p}, r)$;
- phase classification;
- anomaly detection in circulation data;
- computer vision for spectral extraction from coin images;
- reinforcement learning for approximate dynamic policy computation;
- Bayesian optimization over material and geometry parameters.

A supervised phase classifier may estimate

$$\hat{\Pi} = \mathcal{M}_{\theta}(\mathbf{X}),$$

where $\mathcal{M}_{\theta}$ is trained on numerically verified optimization solutions.

The classifier should not replace feasibility checking.

Every proposed design must still satisfy

$$\mathcal{R}(\theta) \geq 0$$

and all engineering constraints.

29 Philosophical Interpretation

The theory distinguishes nominal value from physical form.

A coin of face value F possesses a legally assigned monetary value that need not equal its material value:

$$F \neq p_t M.$$

Its physical body nevertheless affects the cost of supplying the monetary instrument and the incentives to hold or use it.

Thus money has at least two interacting layers:

1. the symbolic and institutional layer;
2. the physical and technological layer.

The geometry of money belongs to the second layer but can influence outcomes in the first. A philosophical interpretation is therefore:

Monetary form is not identical to monetary value,

but

monetary form affects the cost and behaviour associated with monetary value.

30 Religion, Culture, and Symbolic Constraints

Coins often carry symbols with historical, cultural, or religious significance.

Let $\mathfrak{S}$ denote the set of legally and culturally admissible symbolic designs.

The full feasible set becomes

$$\mathfrak{F} = \mathfrak{S} \cap \mathcal{I} \cap \mathfrak{S}.$$

The geometric optimization must therefore coexist with iconographic and institutional constraints.

The framework is neutral regarding any particular religious or cultural symbol. Its purpose is to allow such constraints to enter transparently rather than assuming that geometry alone determines optimality.

31 Strategic and Warfare Economics Interpretation

Coinage can be viewed as a resilience problem.

Let $R_t^{\text{strategic}}$ denote strategic material risk. A simple penalized objective is

$$\mathcal{J}_t^{\text{res}} = \mathcal{J}_t - \lambda_R R_t^{\text{strategic}}.$$

A geometry requiring less scarce material can improve resilience without changing face value.

During disruptions, the authority may face

$$p_t \uparrow, \quad s_t^{\text{supply}} \uparrow, \quad \kappa_t \uparrow.$$

The optimal geometry may then shift toward material conservation or easier domestic production. This interpretation concerns continuity of civilian monetary infrastructure and resource resilience.

32 Vector Graphics

32.1 Constant-width support geometry

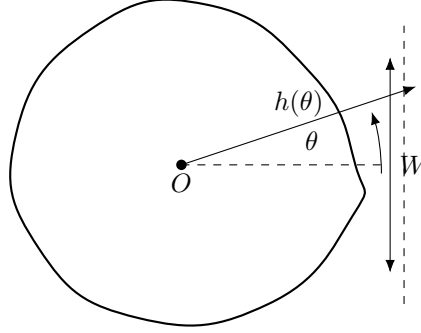


Figure 1: Schematic representation of a non-circular constant-width coin.

32.2 Spectral phase space

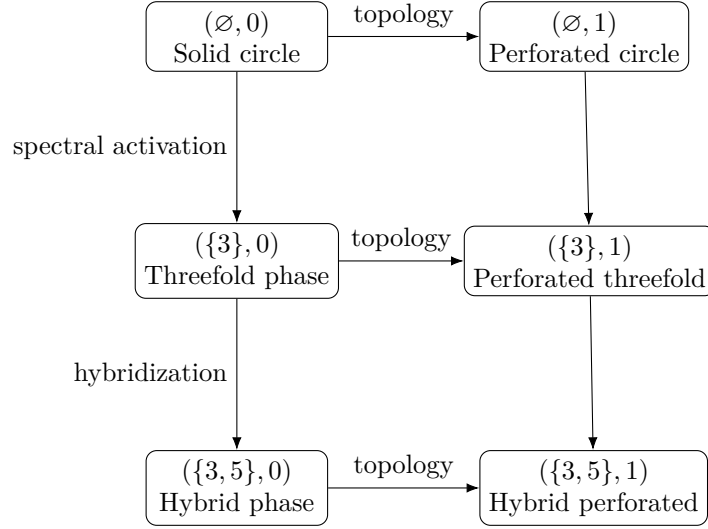


Figure 2: Illustrative spectral-topological phase transitions.

32.3 Dynamic coinage architecture

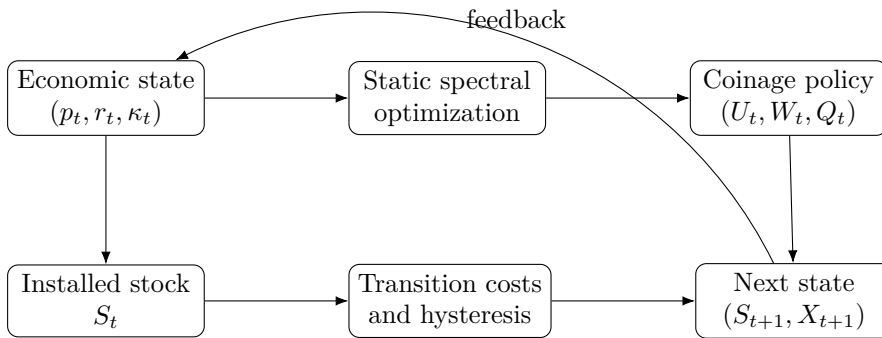


Figure 3: Dynamic architecture of the theory of coinage.

33 Five-Stage Research Program

The complete program is summarized in Table 4.

Table 4: Five stages of the theory

Stage	Object	Main output
I	Single denomination	Constrained optimization over constant-width spectral geometry and topology.
II	Static phase structure	Phase diagram over material prices, interest rates, and technology.
III	Denomination portfolio	Joint geometry allocation with recognition and confusion externalities.
IV	Dynamic installed stock	Bellman equation, transition costs, hysteresis, and replacement policy.
V	Empirical calibration	Structural estimation, state-space inference, and falsifiable hypotheses.

34 Main Predictions

The theory generates the following testable predictions.

1. Rising material prices tend to reduce optimal material fraction:

$$\frac{\partial m^*}{\partial p} < 0$$

under ordinary parameter conditions.

2. Low-frequency odd harmonics tend to activate before high-frequency harmonics when manufacturing complexity rises sufficiently rapidly with frequency.
3. Perforation becomes more attractive when material savings exceed the fixed and variable costs of hole production.
4. Interest rates affect optimal physical design through geometry-dependent cash holding costs.
5. The marginal monetary effect of geometry is greatest near

$$n_c = \frac{1}{2}.$$

6. A multi-denomination system exhibits geometric diversification because recognition is a joint property.
7. Positive transition costs create hysteresis.
8. Strategic material risk can induce redesign even without changes in face values.
9. Improved manufacturing technology can activate previously uneconomic high-frequency geometries.

35 Limitations

Several qualifications are essential.

First, the numerical coefficients introduced above are theoretical benchmark parameters rather than empirical estimates for a particular national mint.

Second, the four-mode truncation is a finite approximation to the full constant-width design space.

Third, the reduced-form logistic cash-demand equation is not a universal law.

Fourth, the relationship between physical geometry and monetary demand must be empirically identified rather than assumed.

Fifth, practical coin design must satisfy manufacturing, accessibility, legal, and public-acceptance constraints that may dominate small theoretical efficiency gains.

Sixth, negative interest rates are an analytically admissible regime but their institutional feasibility depends on the broader monetary system.

36 Conclusion

The theory developed here transforms coinage from a largely exogenous physical form into an endogenous problem in geometry, economics, finance, engineering, and dynamic public policy.

The general monetary instrument is represented by

$$\mathcal{M}_{d,t} = (F_d, W_d, \tau_d, \rho_d, \lambda_{d,t}, \mathbf{z}_{d,t}, \phi_{d,t}, s_{d,t}).$$

The authority's decision is summarized by the policy correspondence

$$\mathbf{Q}_t^* = \Gamma(\mathbf{S}_t, \mathbf{p}_t, r_t, \boldsymbol{\kappa}_t, \mathbf{D}_t, \mathbf{K}_t).$$

Thus optimal coin geometry is a dynamic function of:

$$\boxed{\begin{array}{ccccccc} \text{installed} & + & \text{material} & + & \text{interest} & + & \text{manufacturing} & + & \text{cash} & + & \text{recognition} & + & \text{transition} \\ \text{stock} & & \text{prices} & & \text{rates} & & \text{technology} & & \text{demand} & & \text{requirements} & & \text{costs} \end{array}}.$$

The circle is recovered as

$$\mathbf{z} = \mathbf{0}, \quad \lambda = 0.$$

It is therefore a special equilibrium geometry rather than a geometrically privileged necessity. The broader conclusion is:

$$\boxed{\text{Money can be optimized not only as a financial claim, but also as a physical geometry.}}$$

References

- [1] M. Barbier, *The Geometry of Constant Width*, in standard references on convex and differential geometry.
- [2] T. Bonnesen and W. Fenchel, *Theory of Convex Bodies*, BSC Associates, 1987.
- [3] T. Bonnesen and W. Fenchel, *Theory of Convex Bodies*, translated and edited editions on support functions and constant-width bodies.
- [4] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, Cambridge University Press.
- [5] R. T. Rockafellar, *Convex Analysis*, Princeton University Press.
- [6] P. Krugman, “A Model of Balance-of-Payments Crises,” *Journal of Money, Credit and Banking*.
- [7] M. Friedman, *A Program for Monetary Stability*, Fordham University Press.
- [8] W. J. Baumol, “The Transactions Demand for Cash: An Inventory Theoretic Approach,” *Quarterly Journal of Economics*.
- [9] J. Tobin, “The Interest-Elasticity of Transactions Demand for Cash,” *Review of Economics and Statistics*.
- [10] W. Whitesell, research on currency substitution and the economics of physical currency.
- [11] T. J. Sargent and N. Wallace, “Some Unpleasant Monetarist Arithmetic,” *Federal Reserve Bank of Minneapolis Quarterly Review*.

- [12] R. E. Lucas, “Interest Rates and Currency Prices in a Two-Country World,” *Journal of Monetary Economics*.
- [13] R. Bellman, *Dynamic Programming*, Princeton University Press.
- [14] D. P. Bertsekas, *Dynamic Programming and Optimal Control*, Athena Scientific.
- [15] M. Puterman, *Markov Decision Processes*, Wiley.
- [16] J. D. Hamilton, *Time Series Analysis*, Princeton University Press.
- [17] J. Durbin and S. J. Koopman, *Time Series Analysis by State Space Methods*, Oxford University Press.
- [18] A. Gelman et al., *Bayesian Data Analysis*, CRC Press.
- [19] C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer.
- [20] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press.
- [21] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, MIT Press.
- [22] J. Nocedal and S. J. Wright, *Numerical Optimization*, Springer.
- [23] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press.
- [24] K. J. Arrow, *Social Choice and Individual Values*, Yale University Press.
- [25] A. Sen, *Collective Choice and Social Welfare*, Holden-Day.
- [26] D. C. North, *Institutions, Institutional Change and Economic Performance*, Cambridge University Press.
- [27] D. Kahneman, *Thinking, Fast and Slow*, Farrar, Straus and Giroux.

Glossary

Constant width

A property of a convex body for which the distance between parallel supporting lines is the same in every direction.

Support function

The function $h(\theta)$ giving the distance from the origin to the supporting line with outward normal angle θ .

Spectral mode

An odd Fourier harmonic used to represent deformation away from the circular support function.

Zero-spectrum phase

The state $z_n = 0$ for every non-translational harmonic, corresponding to the circle.

Spectral phase

The set of active harmonic modes together with the coin’s topology.

Topology

The distinction between solid and perforated coin states.

Material fraction

The normalized fraction of the circular reference area remaining after spectral deformation and perforation.

Convexity budget

The limited amount of spectral deformation compatible with nonnegative radius of curvature.

Holding cost

The physical and economic cost of storing, transporting, and using currency.

Monetary geometry window

The interest-rate region in which a marginal change in geometry has its largest effect on cash demand.

Effective lower bound

The geometry-dependent interest-rate threshold at which the attractiveness of physical currency materially constrains further rate reductions.

Geometric diversification

Joint selection of different coin geometries across denominations to reduce recognition and confusion costs.

Installed stock

The existing distribution of coins in circulation.

Transition cost

The economic cost of replacing one coinage geometry with another.

Hysteresis

Dependence of the current optimal or observed state on the history of previous states and switching costs.

Phase diagram

A map assigning an optimal spectral-topological state to each point in an economic parameter space.

Strategic material risk

Additional economic cost associated with supply concentration, disruption, or other threats to material availability.

The End